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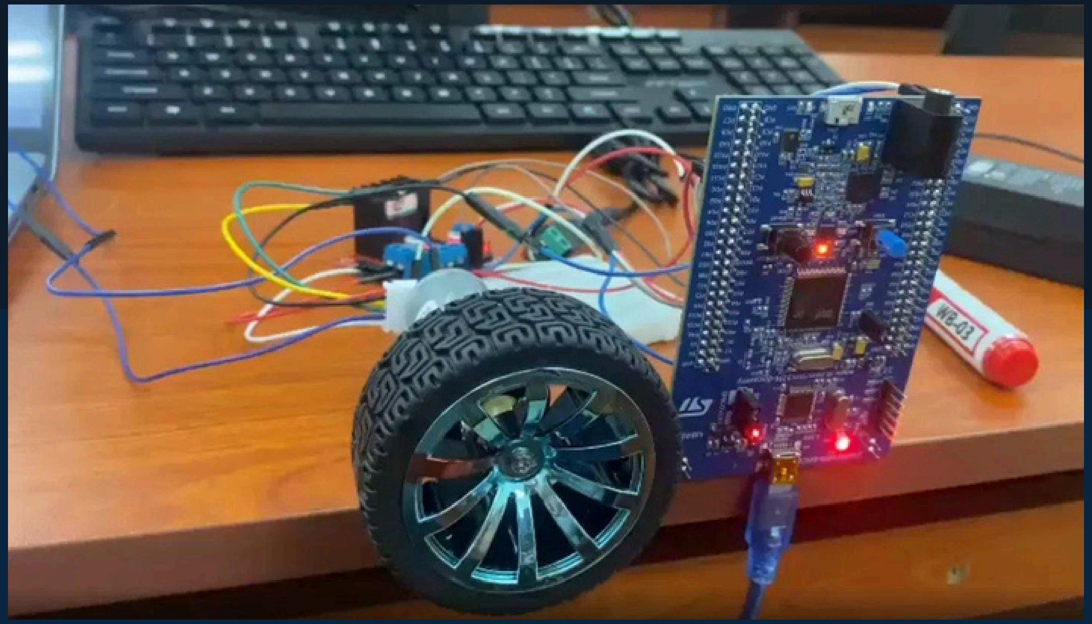
Design and Implementation of a Motion Control System

Using DC Motor and STM32 Microcontroller



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School of Electrical Engineering | June 2026

Presentation Outline

01 Introduction & Motivation

Background

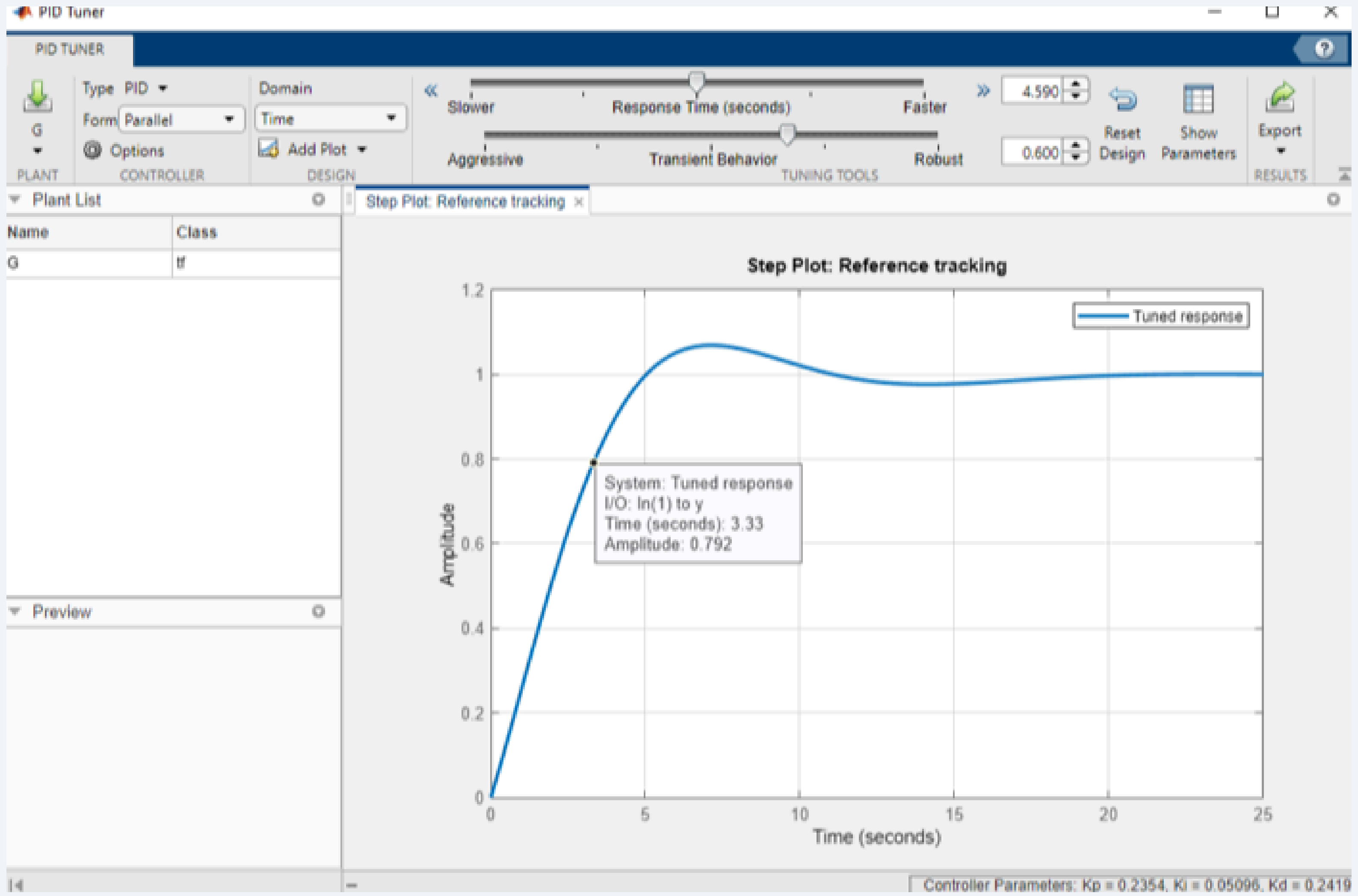
PID control is used in >90% of industrial control loops worldwide. Modern embedded MCUs enable model-based digital control design.

Problem Statement

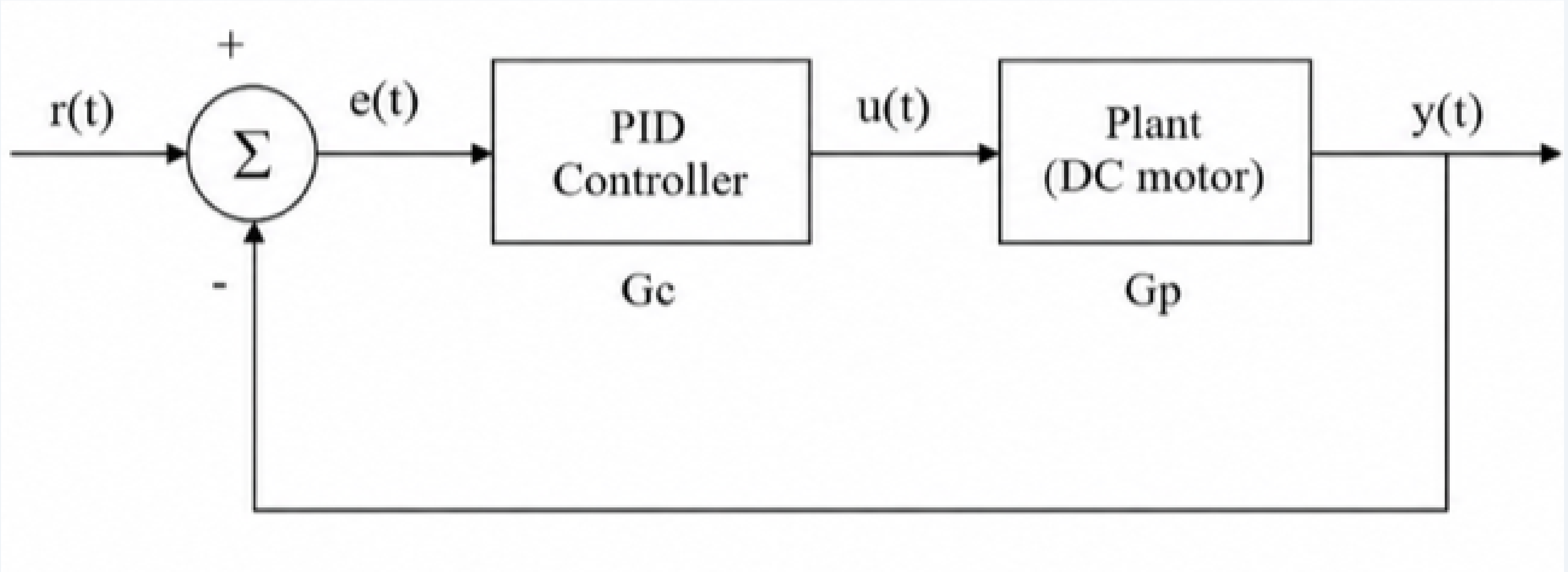
Achieve precise, repeatable 90° positioning (≈515 encoder counts) on a DC motor with minimal overshoot & fast settling — despite nonlinearities like friction, PWM deadzone, and sensor noise.

Project Objectives

- Assemble hardware prototype (STM32F407 + DC motor + L298N)
- System identification via open-loop experiment + MATLAB
- Automated PID parameter optimization using pidtune()
- Validate closed-loop performance with real-time telemetry

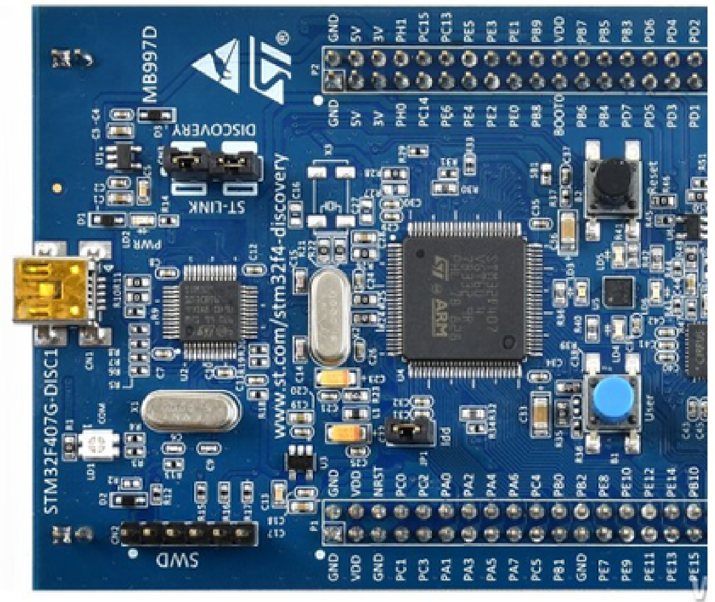


PID Tuner - MATLAB



Closed-Loop Control Concept

02 Hardware Components



STM32F407VG Discovery

168 MHz ARM Cortex-M4
Main processing unit

$15

DC Motor + Encoder

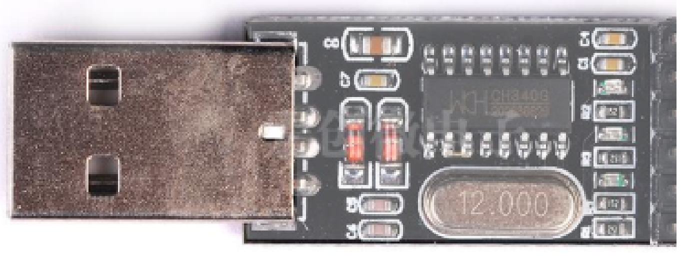
12V, 2060 pulses/rev
Quadrature feedback

$8

L298N H-Bridge

Dual full-bridge driver
Bidirectional motor control

$2.50



CH340 USB-UART

Serial telemetry
Real-time monitoring

$2

Total Project Cost: $38.50 | MATLAB Student License: Free | Tools: STM32CubeIDE + Teleplot

02 System Architecture Overview

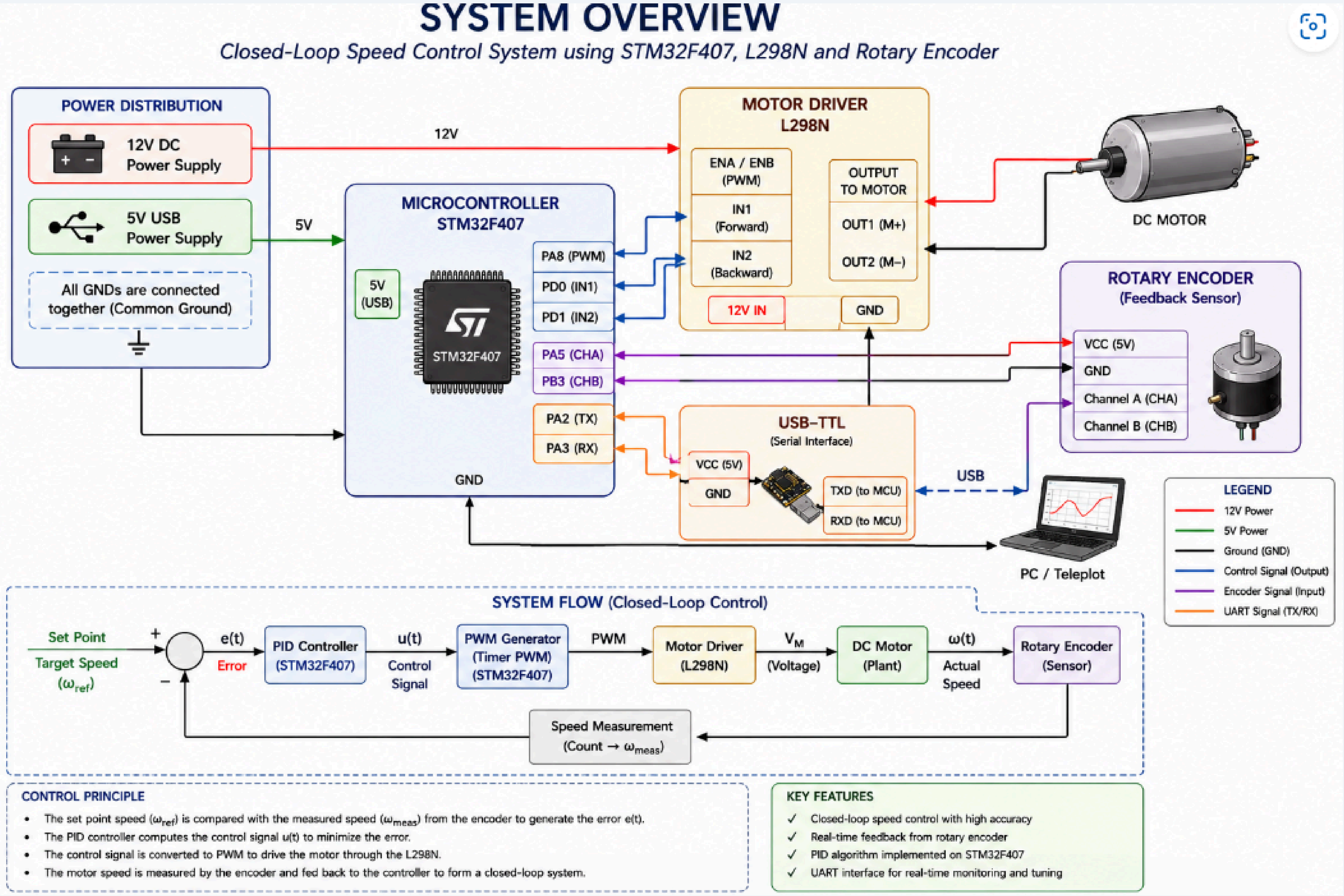
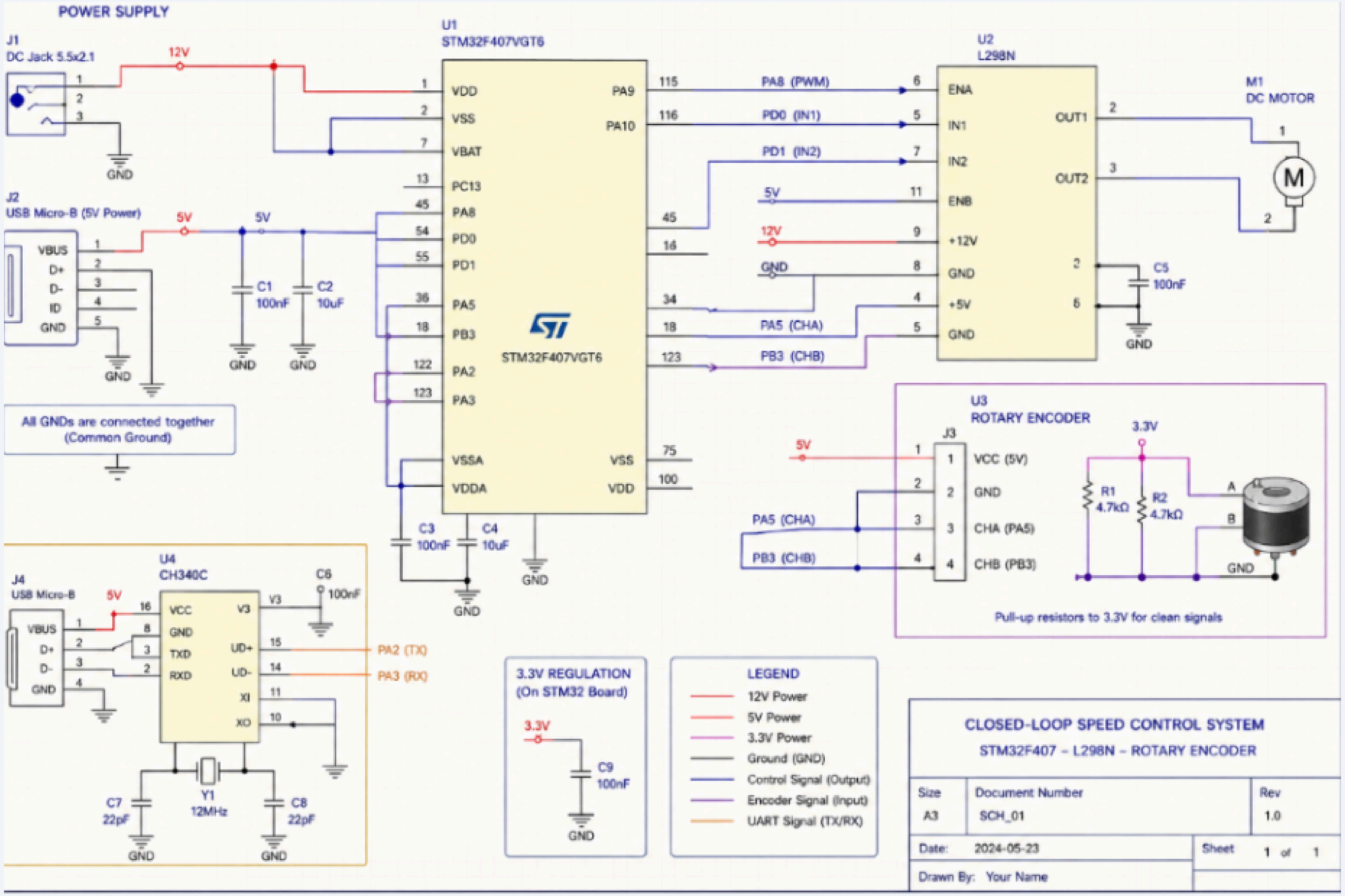


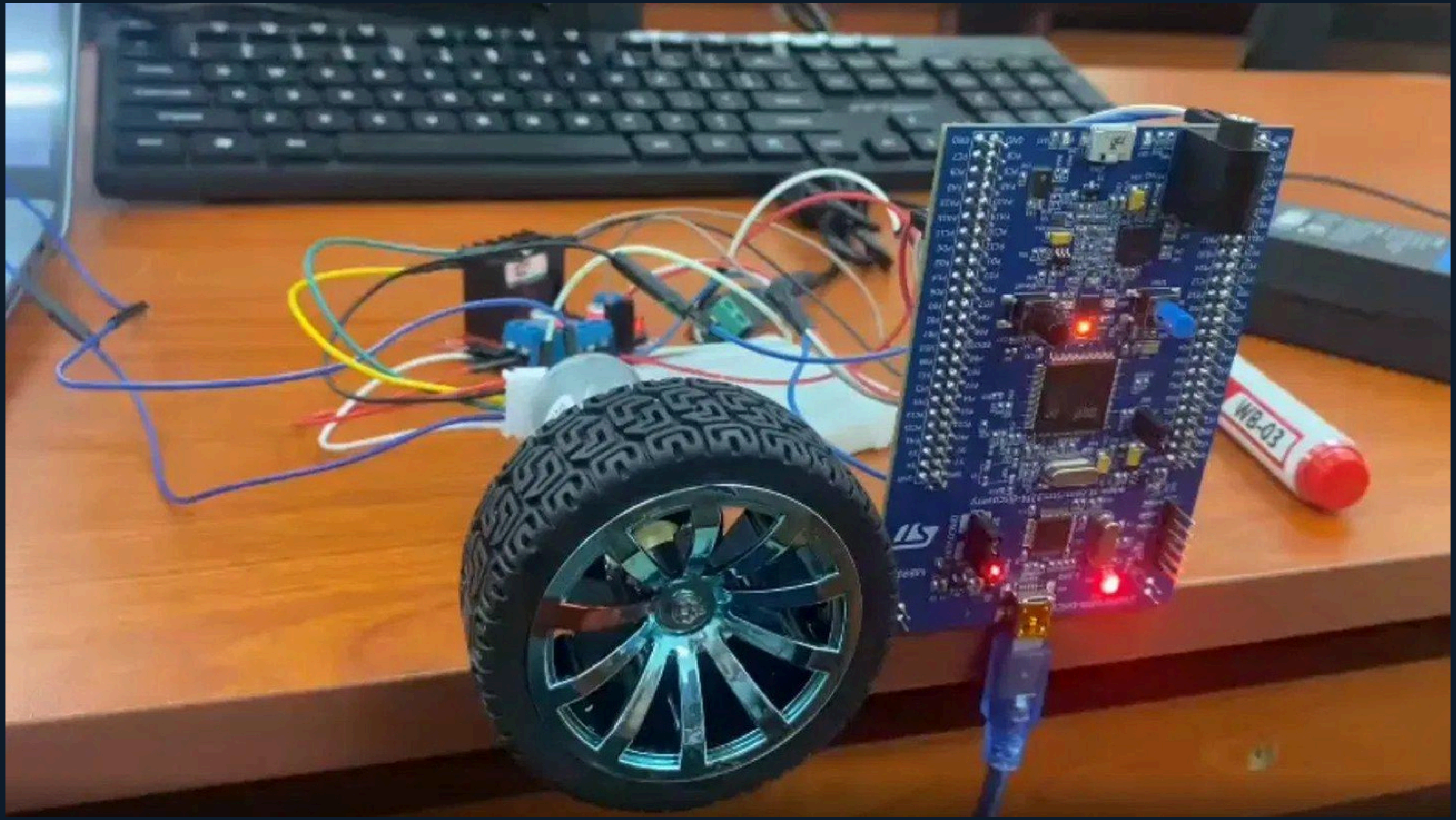
Figure: Full System Schematic with Signal Flow

02 System Architecture Overview

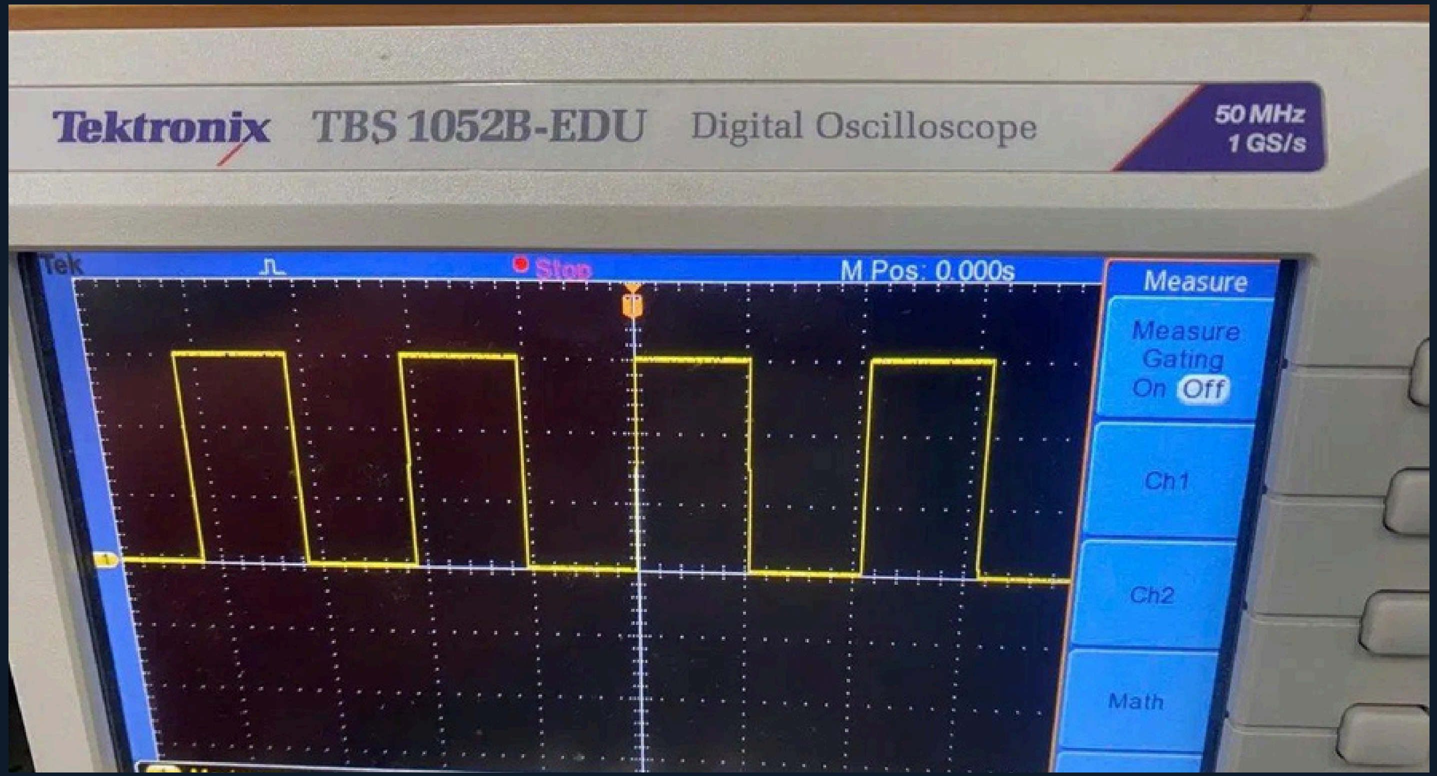


Circuit Schematic (STM32 + L298N + Encoder)

Hardware Prototype & Signal Verification



Physical Prototype: STM32F407 + DC Motor + L298N on breadboard



PWM Signal: 3.04 V, 50% Duty, 95.78 Hz

Key Design Points

TIM1 → PWM on PE9 (20 kHz)

TIM2 → X4 Encoder (PA5, PB3)

4.7 kΩ pull-ups to 3.3 V rail

USART2 @ 115200 baud (PA2/PA3)

03 Methodology: Mathematical Modeling

Mechanical (Newton): $T = J \cdot d\omega/dt + b \cdot \omega$

Laplace → Position Transfer Function:

$$G(s) = \Theta(s)/V(s) = Kt / [s \cdot (Ls+R) \cdot (Js+b) + KtKe \cdot s]$$

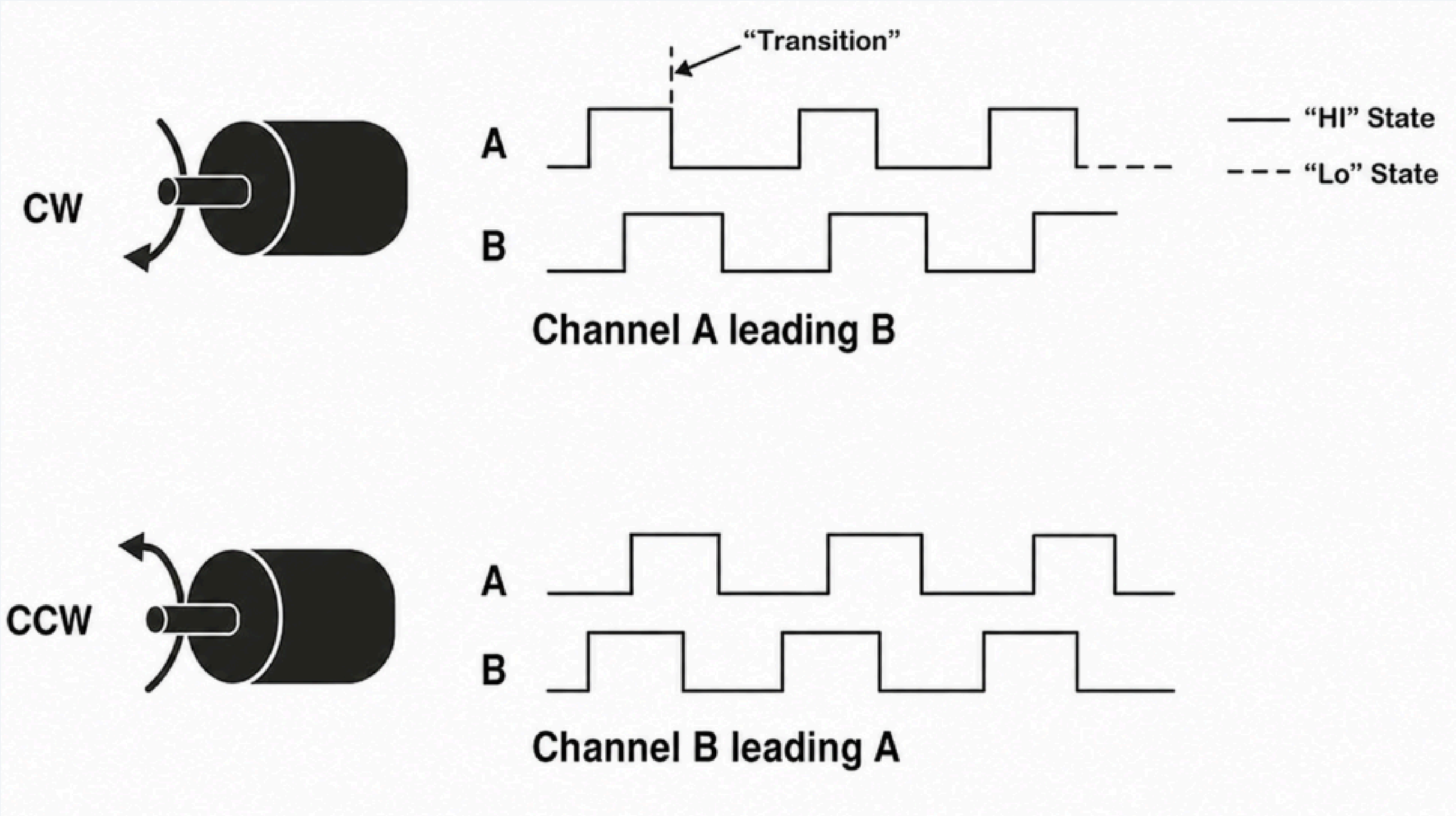
→ Contains an integrator (1/s) + 2 poles

→ TF structure: TF(2 poles, 0 zero)

Quadrature Encoder (X4 Mode):

Total Pulses = PPR_base × 4 × Gear_ratio

= 2060 pulses per revolution



X4 Quadrature Decoding – counts all edges of both channels

Identified Plant Model:

$$G(s) = 8.9958 / (s^2 + 3.9716s + 2.531 \times 10^{-4})$$

→ 99.51% validation fit

03 System Identification with MATLAB

1

Data Collection

Open-loop PWM=450
84 data points over 0.83 s via UART

2

Preprocessing

Clean duplicates
60/40 train/validation split

3

tfest() in MATLAB

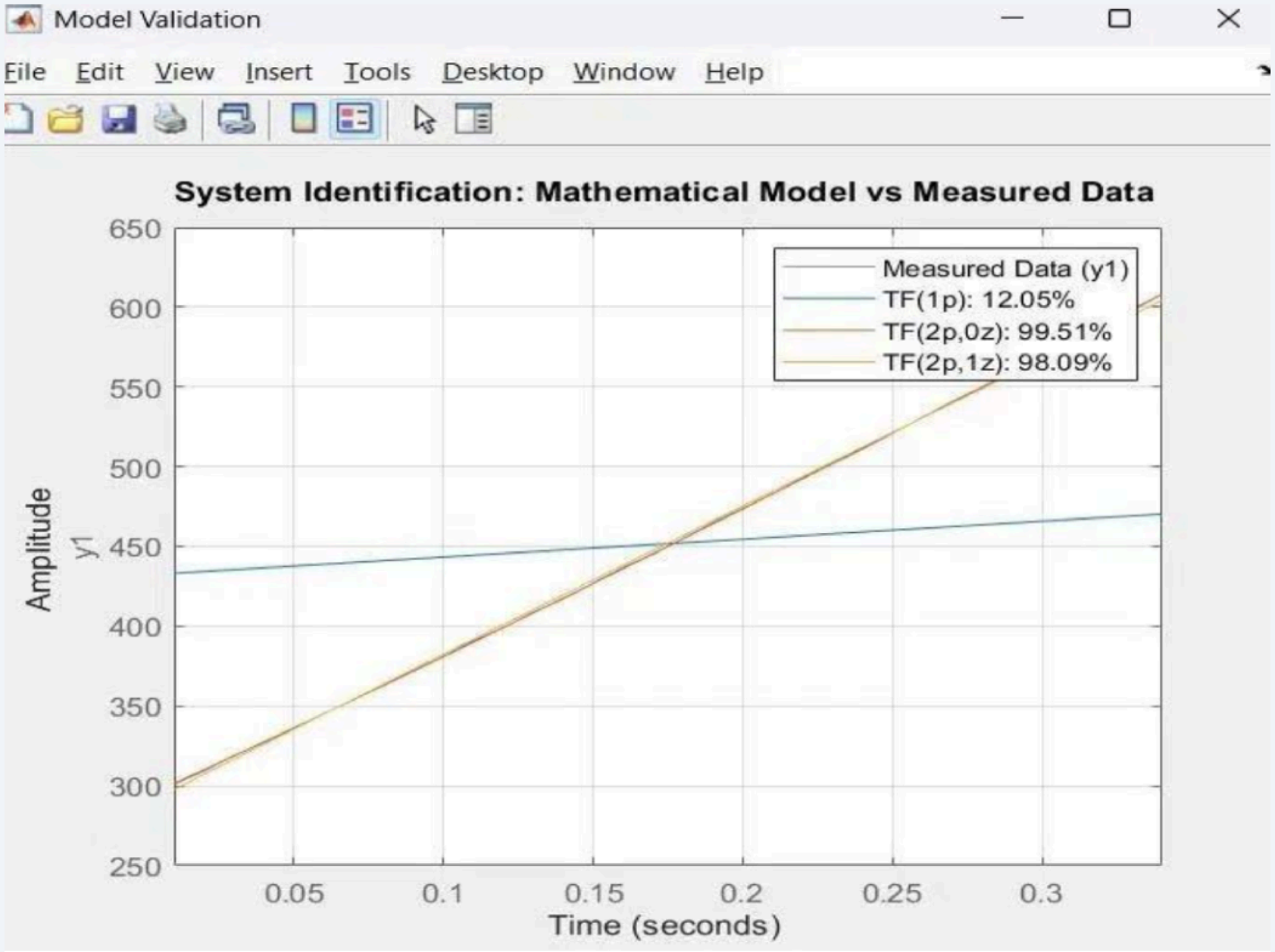
Compare TF(1p), TF(2p,0z), TF(2p,1z)

4

Model Selected

TF(2p,0z): 99.51% fit
→ Best accuracy & simplicity

03 System Identification with MATLAB



MATLAB System Identification: Model vs Measured Data

Model Comparison Results

TF(1 pole, 0 zero):

12.05% fit — REJECTED

TF(2 poles, 0 zero):

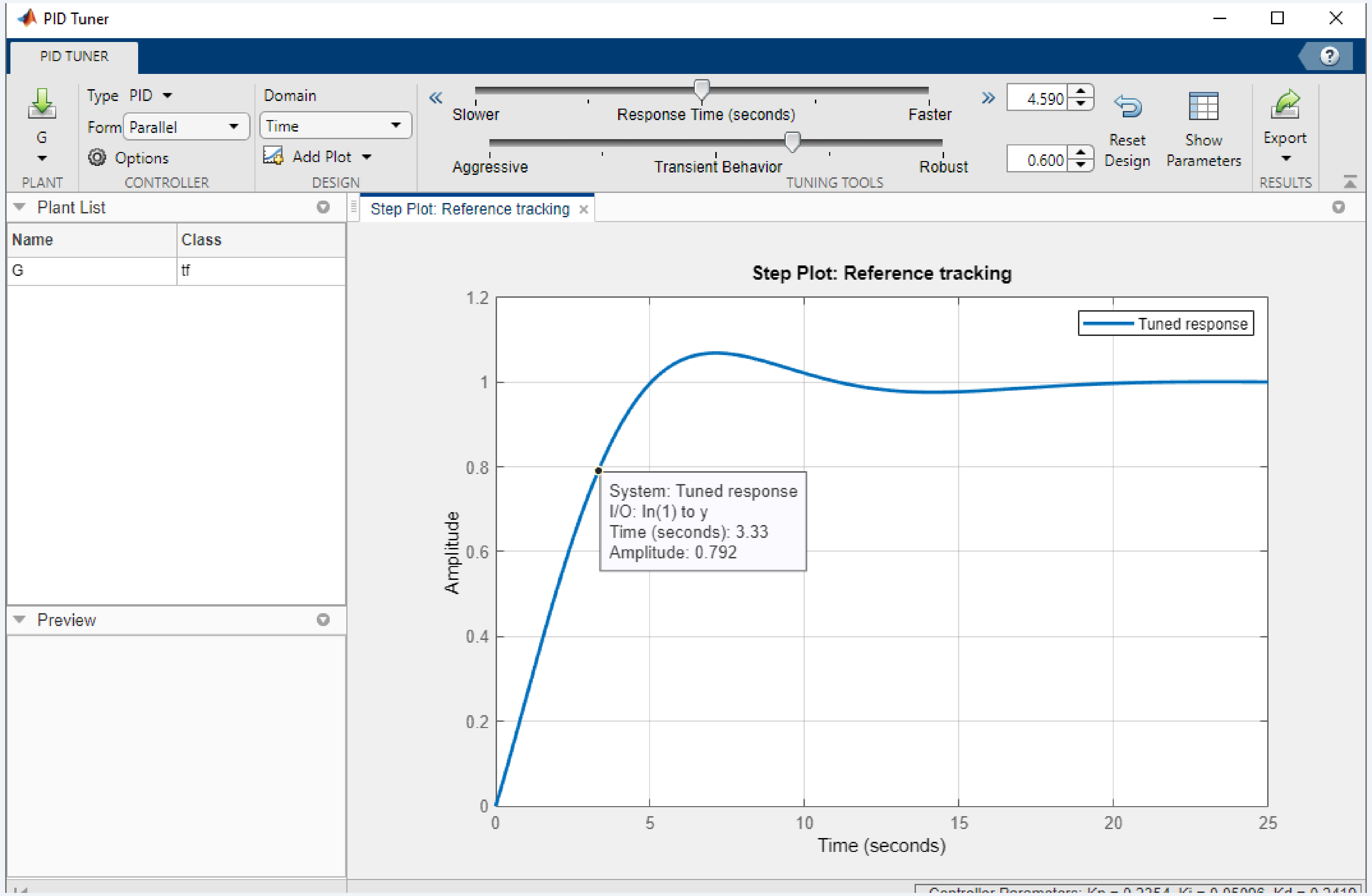
99.51% fit — ✓ SELECTED

TF(2 poles, 1 zero):

98.09% fit — Good, but unnecessary complexity

$$G(s) = \frac{8.9958}{s^2 + 3.9716s + 2.531\times10^{-4}}$$

03 PID Controller Design & Discretization



MATLAB pidtune() Grid Search: ωc = 8 rad/s, PM = 65°

Discrete PID (Forward-Euler, Ts = 10 ms)

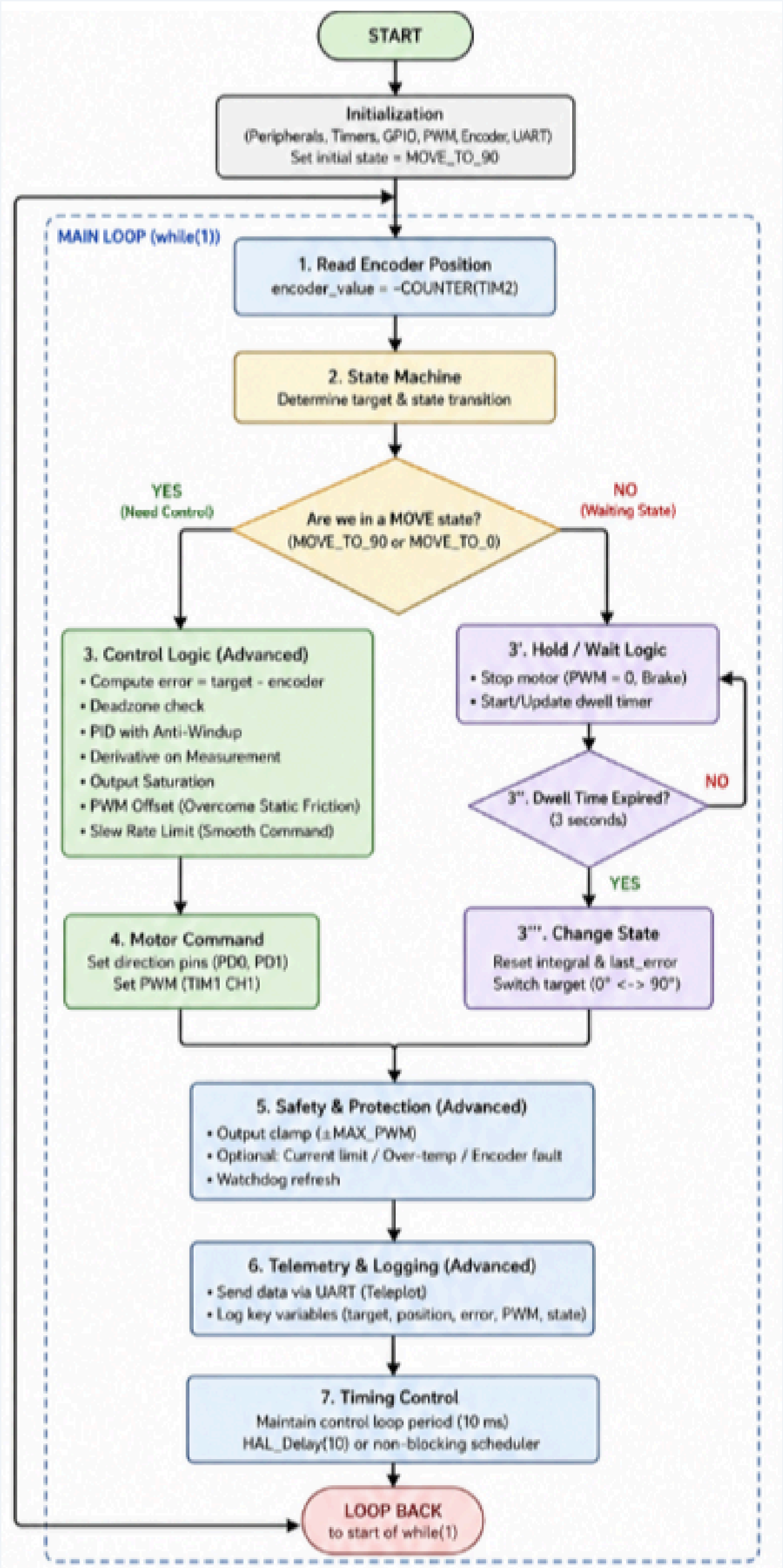
u[k] = Kp·e[k] + (Ki·Ts)·Σe + (Kd/Ts)·(e[k]−e[k−1])

⚠ Derivative Noise: Kd/Ts = 0.9117/0.01 = 91.17

→ Kd manually adjusted to 3.0 (noise suppression)

Parameter	Continuous (MATLAB)	Discrete (Firmware)
Kp	4.1671	4.1671 (unchanged)
Ki	4.2523	0.0425 (× Ts)
Kd	91.17	**3.0 (adjusted)**
Settling Time	1.897 s (sim)	**≈ 1.90 s (hw)**
Overshoot	9.3%	**≈ 9.3%**

03 Embedded Software: FSM & Control Logic



FSM Flowchart

MOVE_TO_90

Target = 515 counts

PID active, motor running

WAIT_AT_90

Error ≤ ±6 counts

Hold 3 s, clear integral

MOVE_TO_0

Target = 0 counts

PID active, reverse drive

WAIT_AT_0

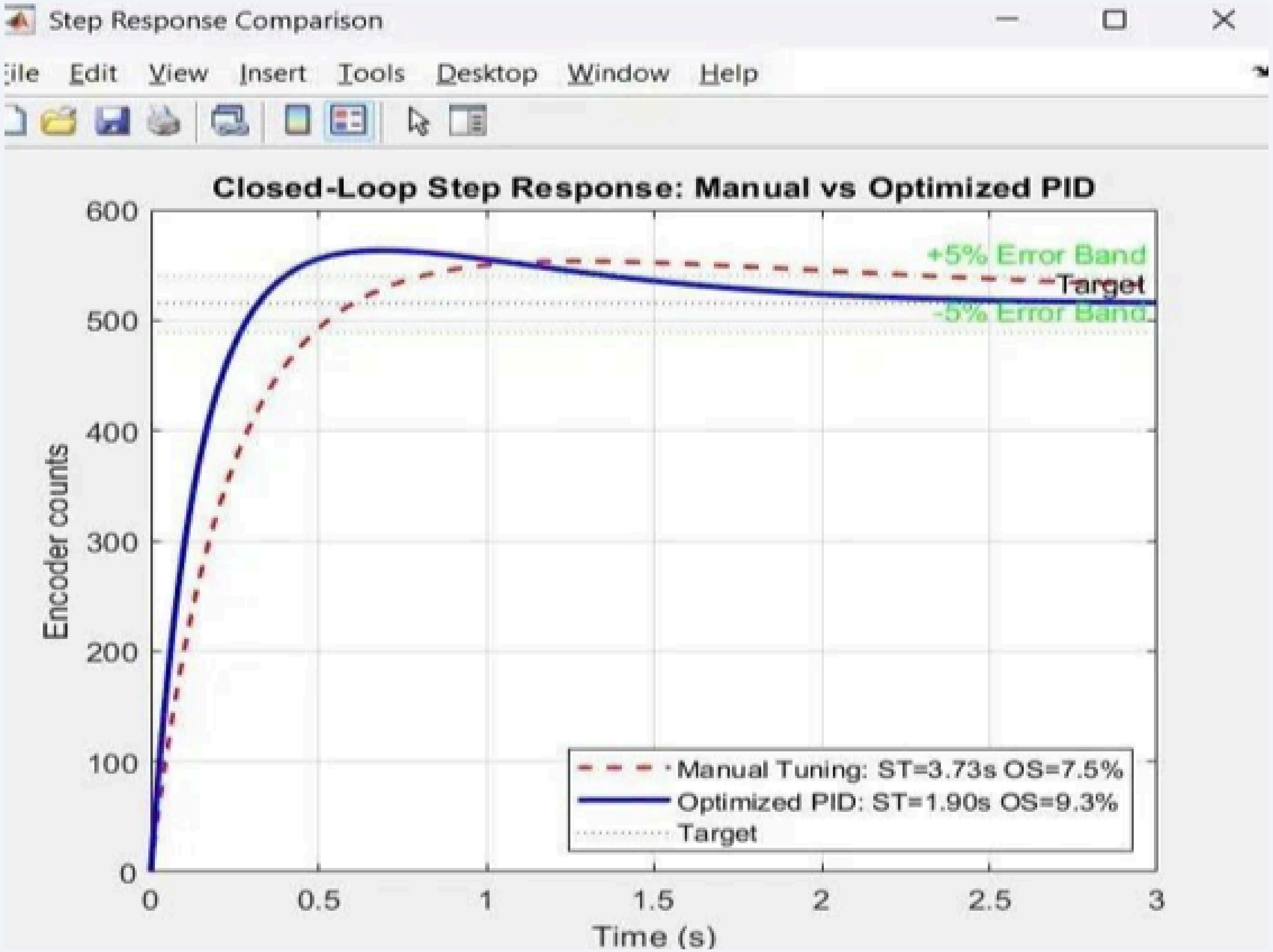
Home reached

Hold 3 s → repeat cycle

Advanced Control Features:

Anti-Windup: integral accumulates only when |error| < 100 | **Deadzone:** PWM cut to 0 when |error| ≤ 6 counts | **Stiction Offset:** +40 PWM counts to overcome gearbox friction

04 Results: Simulation Step Response



MATLAB Closed-Loop Step Response: Manual vs. Optimized PID

Performance Comparison

Settling Time

Manual: 3.73 s Opt: 1.90 s

Overshoot

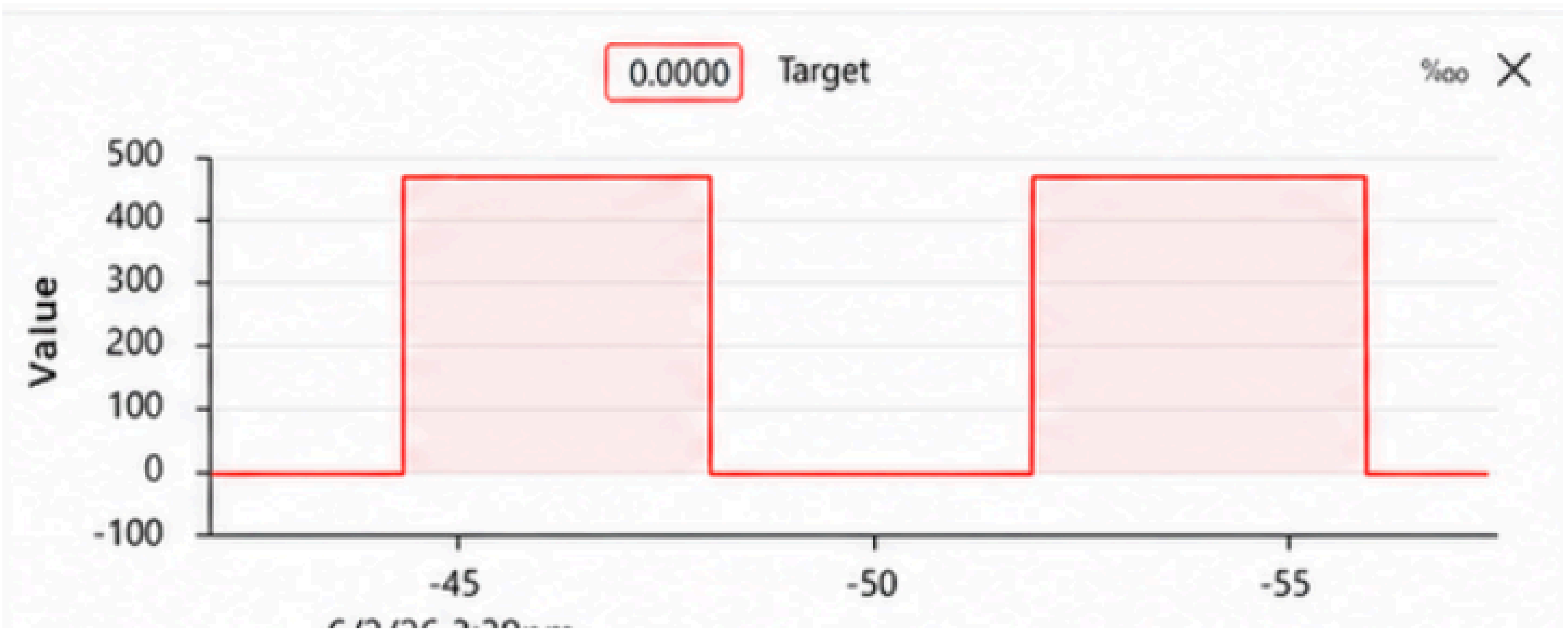
Manual: 7.5% Opt: 9.3%

Steady-State Error

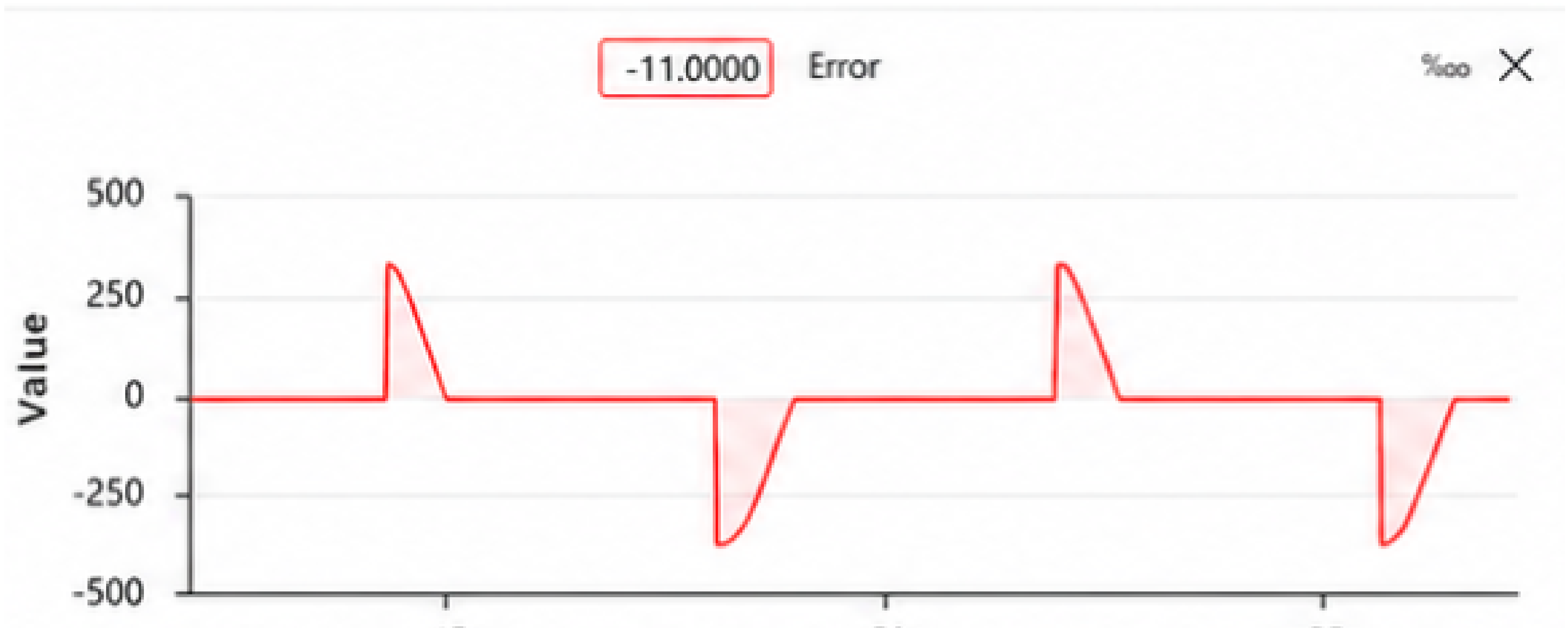
49% faster settling time
(3.73 s → 1.90 s)

04 Results: Real-Time Hardware Performance

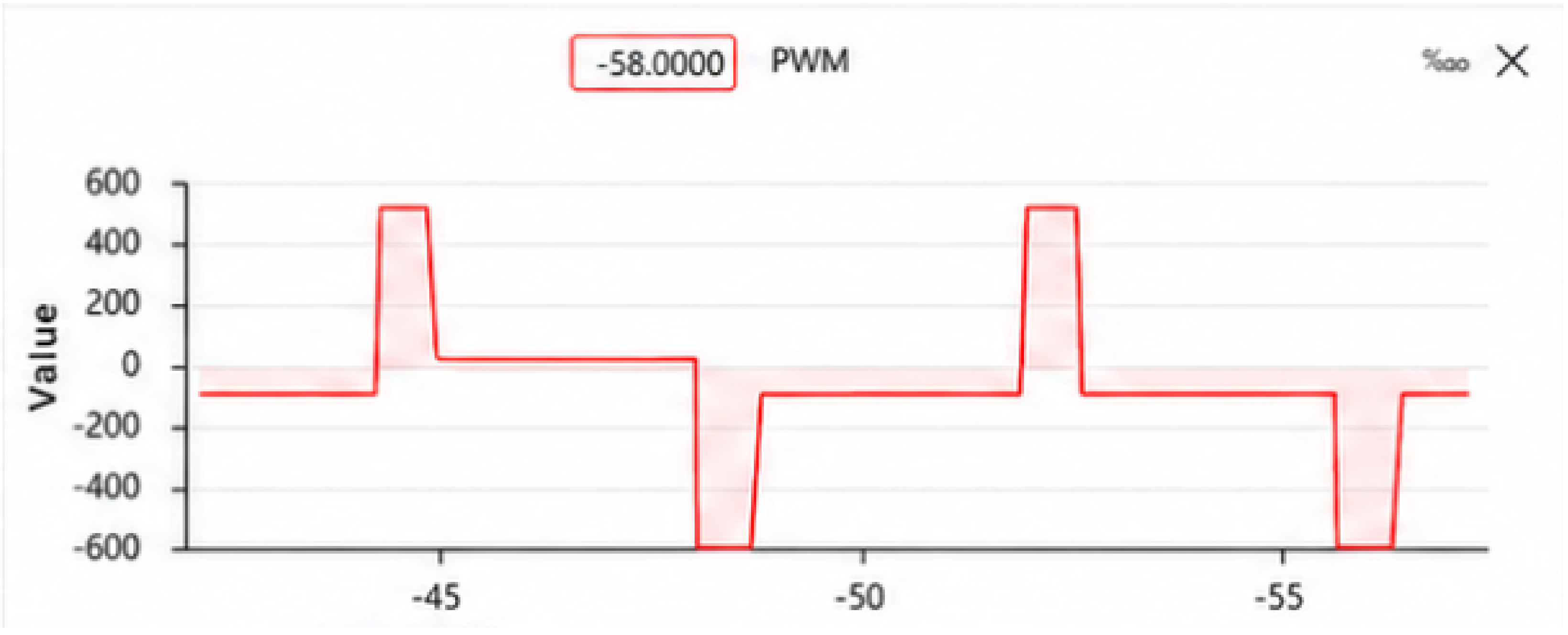
Encoder Position (0 ↔ 515 counts)



Target Position (Step Input)



Position Error (spike then → 0)



PWM Output (bidirectional drive)

Steady-state error within ±6 encoder counts | Stable 0°→90°→0° cycling confirmed | Bidirectional PWM control verified

04 Discussion: Key Engineering Decisions

D1

Model Accuracy & System ID

Problem: Unknown motor parameters (R, L, J, b) | **Solution:** Open-loop step experiment + MATLAB tfest() | **Result:** TF(2p,0z) = 99.51% fit — validates double-integrator behavior

D2

Derivative Noise Amplification

Problem: Theoretical Kd_code = 91.17 caused severe oscillation | **Solution:** Manually reduced Kd to 3.0 (per MATLAB warning) | **Result:** Stable operation; encoder quantization noise suppressed

D3

PWM Deadzone & Stiction

Problem: Motor stalls for small errors due to gearbox friction | **Solution:** PWM offset +40 + DEADZONE ±6 counts | **Result:** Reliable positioning; no limit-cycling at setpoint

07 Conclusion & Future Work

Conclusions

- Hardware prototype assembled and operated reliably
- $G(s) = 8.9958/(s^2+3.9716s+2.531\times10^{-4})$ with 99.51% fit
- Optimized PID: Kp=4.1671, Ki=0.0425, Kd=3.0
- 49% faster settling: 3.73 s → 1.90 s (9.3% OS)
- Stable 0°↔90° cycling validated via Teleplot
- Complete model-based design workflow demonstrated

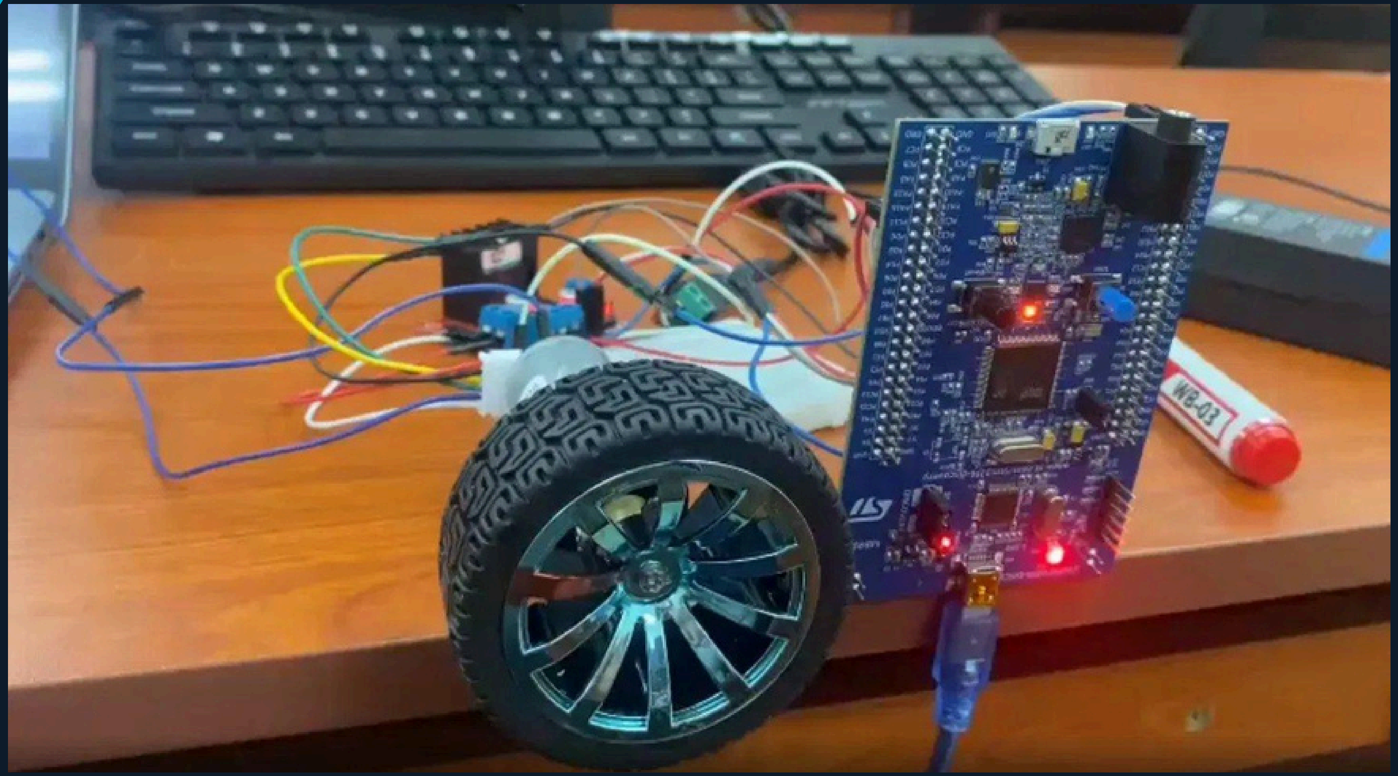
Future Work

- Custom PCB to reduce EMI
- Trapezoidal motion profiling
- LQR or Fuzzy Logic control
- Derivative low-pass filter
- optimal control for electrical car

Successfully demonstrated a complete model-based digital PID control workflow:
Hardware → System ID → Controller Design → Simulation → Hardware Validation

Thank You

Questions & Discussion



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